STORAGE

Panel: Comprehensive Validation Testing of OCP Datacenter NVMe SSD Specification Requirements

Tim Sharp, *Principal Hardware Engineer – Microsoft Corporation*

Rick Walsh, *Sr. Vice President – SANBlaze*

Nick Kriczky, *Vice President of Products and Services – Teledyne LeCroy*

OCP PLATINUM Member

Microsoft

STORAGE

Driving an Open-Source Approach to Datacenter NVMe SSD Requirements Validation

OCP PLATINUM Member

Open standard: *OCP Datacenter NVMe SSD Specification (DSSD)*

Shared test suite: C*ommon test cases and methodology across the industry*

Multi-vendor: *implemented by SANBlaze and Teledyne LeCroy platforms*

Microsoft at the 2024 OCP Global Summit

- Introduced the work being performed by Microsoft to develop device level tests to verify SSD support for testable Datacenter NVMe SSD Specification requirements
- Described Microsoft's process of test case and test plan development used by and the common format of test cases
- Introduced the engagement Microsoft had begun with third party tool vendors Teledyne LeCroy and SANBlaze that each implement the DSSD test cases defined by Microsoft
- Described the specific benefits to the SSD industry from adopting and supporting the open standard defined by the OCP Datacenter NVMe Specification

Since the 2024 OCP Global Summit

- Microsoft has continued to create and update test cases for each subsequent releases of the OCP Datacenter NVMe SSD Specification
  - OCP Datacenter NVMe Specification 2.5 (183 tests defined)
    - 48 new OCP test cases were added
    - Many existing test cases were updated to support new or updated requirements
  - OCP Datacenter NVMe Specification 2.6 (191 tests defined)
    - 8 new test cases have been added
    - Many existing test cases have been updated to support new or updated requirements
    - Test development work continues
  - OCP Datacenter NVMe Specification 2.7
    - Gap analysis of OCP2.7 is complete
    - Test case development started 9/2026
    - 6-10 new test cases will be added
    - Many existing test cases will be updated to support new or updated requirements

Since 2024 OCP Global Summit (continued)

- Comprehensive Test Plan for OCP Datacenter NVMe SSD Specification Testing
  - To be contributed to OCP by Q1 2027
- Third Party Tool Vendor Engagement
  - Continued the close engagement with Teledyne LeCroy and SANBlaze to roll out new test suites to validate each of the latest OCP spec releases
  - Continued to turn supplier and vendor feedback to bug fixes and enhancements across OCP2.0/2.5/2.6 test suites
  - Worked with tool vendors to add new test tool logic to increase efficiency and reduce test suite execution time:
    - Logical decomposition of test cases into groups based on dependencies to allow suite to be executed on separate platforms in parallel to reduce over all execution time
    - New preconditioning group feature added to allow a single preconditioning phase to be performed and applied to all test cases that require preconditioning for a steady state requirement
    - Requesting tool vendors to add support for the SPRandom preconditioning algorithm as a replacement for conventional preconditioning

Since the 2024 OCP Global Summit (continued)

- Added Support for Error Injection
  - Released Error Injection white paper defining expected device behavior for the original 10 panic types defined in OCP2.0
  - Developed new error injection test cases and added to OCP2.0/2.5/2.6 test suite definitions
    - CPU/Controller Hang Test Case
    - NAND Hang Test Case
    - PLP Defect-Read Only Mode Test Case
    - PLP Defect-Write Through Mode Test Case
    - PLP Defect-Normal Mode Test Case
    - Logical Firmware Error Test Case
    - DRAM Corruption Critical Path Test Case
    - DRAM Corruption Non-Critical Path Test Case
    - NAND Corruption Test Case
    - SRAM Corruption Test Case
    - HW Malfunction Test Case
    - No More NAND Spares-Read Only Mode Test Case
    - No More NAND Spares-Write Through Mode Test Case
    - No More NAND Spares-Normal Mode Test Case

Since the 2024 OCP Global Summit (continued)

- Added New Error Types to Error Injection
  - OCP2.5
    - Incomplete Shutdown error
  - OCP2.6
    - Metadata Corruption error
    - Critical Garbage Collection error
    - Latency Spike error
    - I/O command failure error
    - I/O command timeout error
    - Admin command failure error
    - Admin command timeout error
    - Thermal Throttle Engaged error
    - Thermal Throttle Disengaged error
    - Critical Temperature error
    - NAND Die Offline error



Since the 2024 OCP Global Summit (continued)

- Added New Event Types to Event Injection
  - OCP2.7
    - Error Injection Renamed Event Injection, Panic ID is now Event ID
    - Substantial update defining expected device behavior during error conditions
    - Added 10 new Event Injection types
  - OCP2.8
    - Even more refinement of expected device behavior during error conditions
    - Error Recovery Workflows substantially updated
    - Added 7 new Event Injection types



What’s Coming in 2027?

- Continue to Develop and Modify Tests for OCP2.8 and Beyond
- Develop OCP Datacenter NVMe SSD Specification 2.9
- Continue the Third-party Tool Developer Engagement
- Contribute the Comprehensive OCP Test Plan to OCP



Call to Action – Help Us Build an Open-Source Industry Standard for OCP SSD Testing

Microsoft Is Working To Enable Consistent, Repeatable Testing Of Devices Implementing OCP SSD Requirements

We Invite The Industry To Collaborate On:

- OCP and NVMe requirements interpretation
- Common test methodologies and test development
- Alignment on expected device behavior
- Sharing feedback and lessons learned

Let’s Build the Test Standard Together.

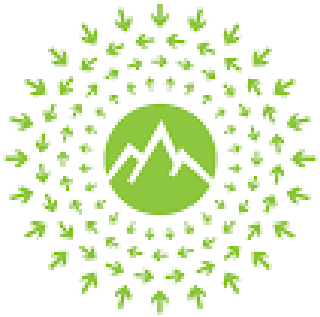


SANBlaZe

Rick Walsh, Sr. Vice President



OCP ACCEPTED™



OCP GLOBAL SUMMIT

October 2026

OCP Standards Drive Consistency

- Aligns SSD requirements between Hyperscale/OEMs and SSD makers
  - AI becomes more important and targeted applications
- Contributes to shared learnings based on Hyperscale and Enterprise configurations
- Enables focus on “customer” requirements
  - More efficient SSD Development – saves money
  - Enables 3rd party test development
  - Aligns drive vendors on specific cloud-based SKU’s
  - More Features, Better Quality, and Faster Delivery

TEST & VALIDATION



STORAGE



SANBlaZe

Storage Test Consistency

- Benchmark level of Data Center Drive Metrics
  - While still not an officially sanction program or certification
  - SANBlaze Offers a test methodology to insure OCP standards and requirements are met
  - Supporting the Microsoft Test Plan Requirements for 2.6 and 2.7 and Beyond
  - Industry adoption!

- Hardware Platforms
  - RM6- 16 drive solution/Scaled Testing
  - DT5- 4 drive Solution
  - IP is the same across both platforms

SBExpress Hardware OCP Test Platform

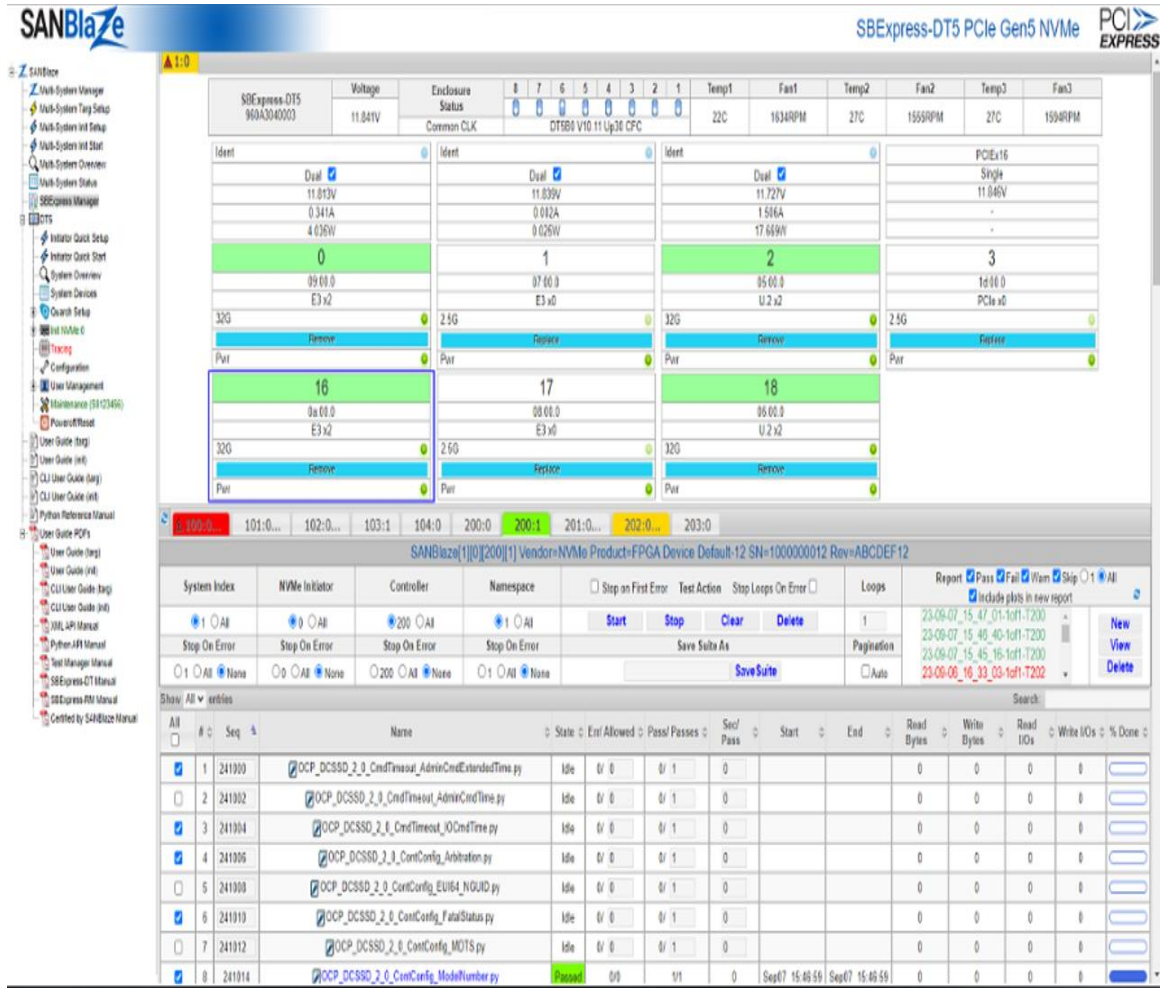




- Rackmount 16 Drive Solution, all front accessible
- Automated OCP Scaled Testing
- SANBlaze IP
  - NSSR Testing
  - Power Margining
  - Power Substate Control
  - 12.5 MHz clocking MI support
  - I3C support
- AIC x16 add in card slot available /FPGA test support

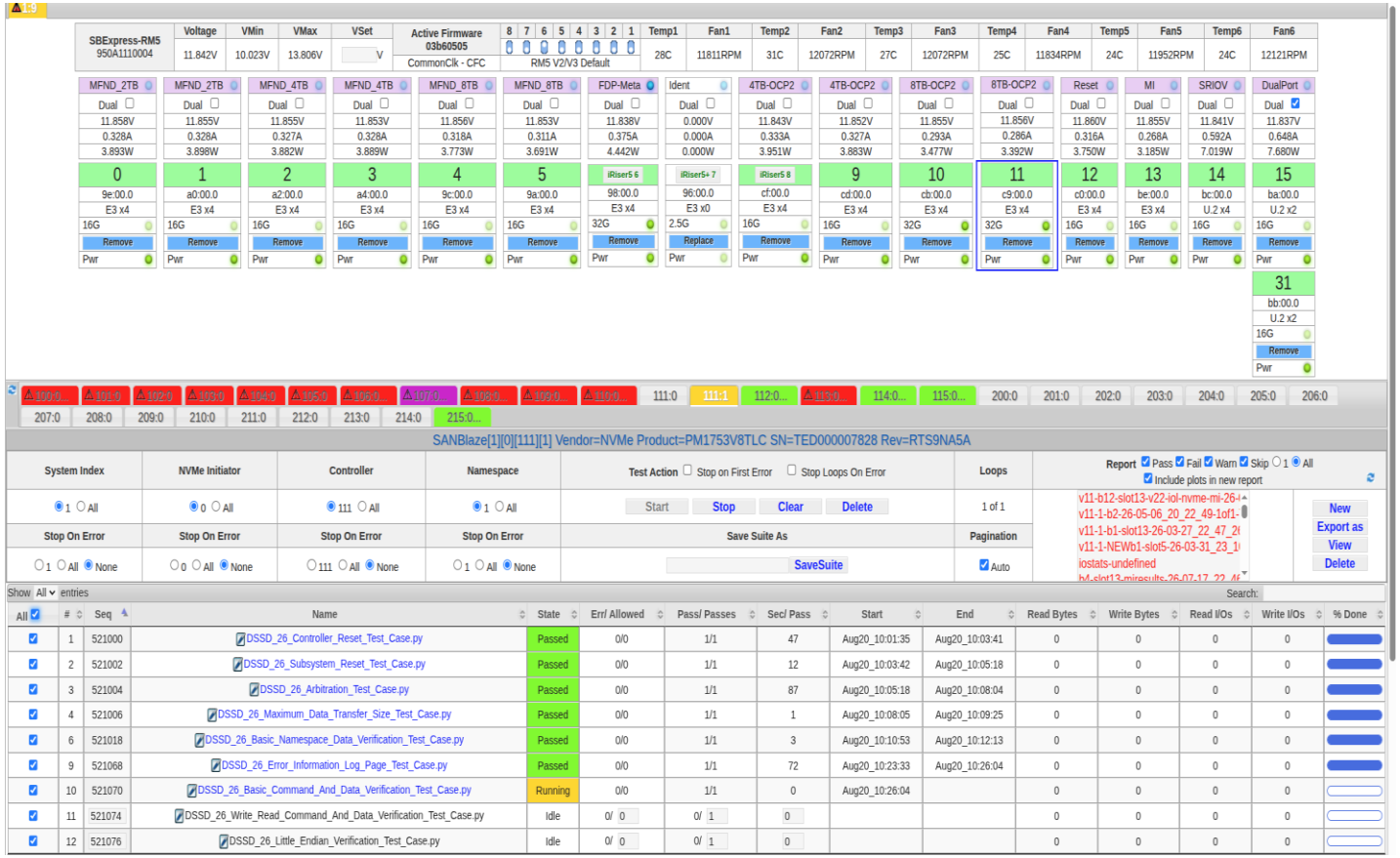
OCP Test Coverage and Infrastructure

- OCP 2.6 and 2.7 Target Test Platforms
- for Data Center SSD's
- Hardware supports up to 16
  - Single/Dual Port SSD's
  - Desk top Platform
- OCP Test Plan Requirements
  - Automated Test Suite
- Follows full Test Plan Requirements as
- specified by Microsoft
  - Data Center 2.6/2.7 spec for SSD's



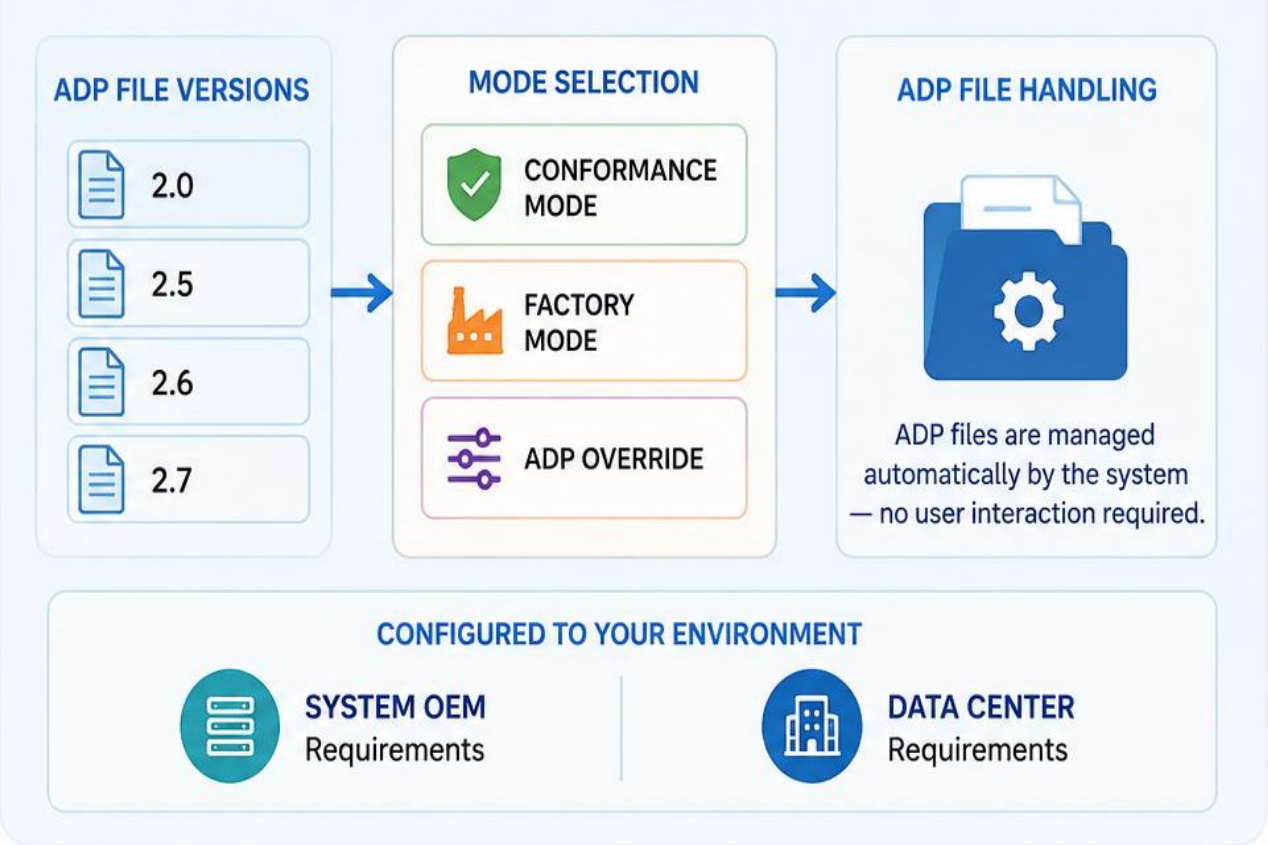
OCP Test Coverage and Infrastructure

- Automated OCP Test Suite:
  - Certified by SANBlaze
- Microsoft CDL Approved
- UNH-IOL Conformance
  - License Available
- Python API support
  - Customize your own tests
- 2.7- Q1 2027



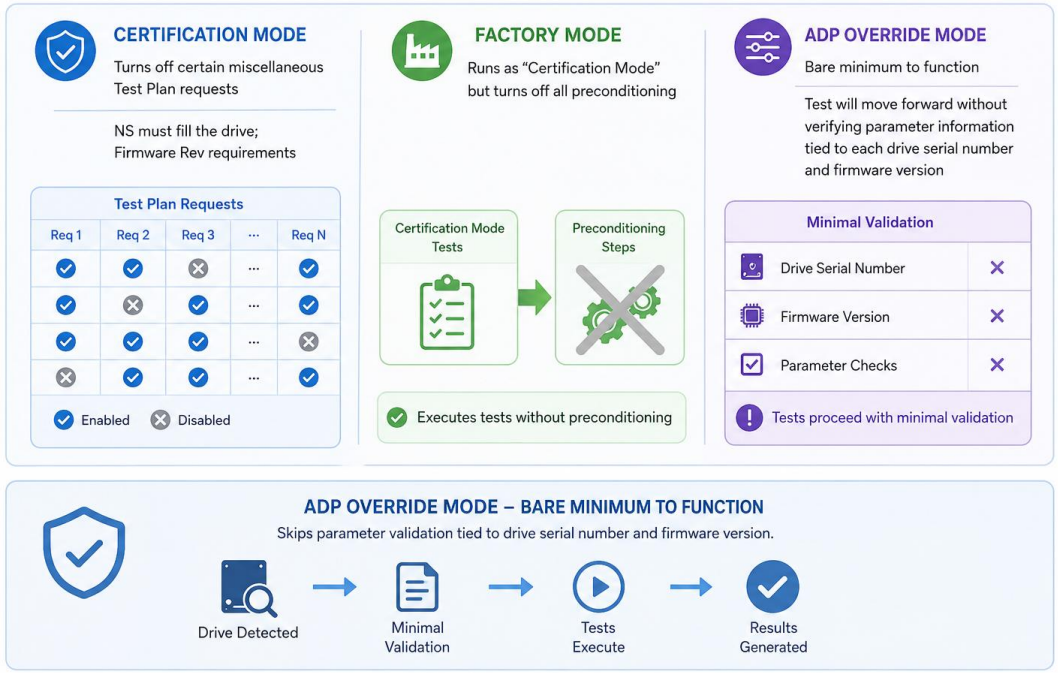
ADP Optional File Management

- Introduction ADP File Management:
  - 2.0/2.5/2.6/ 2.7
  - **Conformance Mode**
  - **Factory Mode**
  - **ADP Override**
- Allows for ADP file support as “optional”
- Select System OEM and Datacenter Requirements



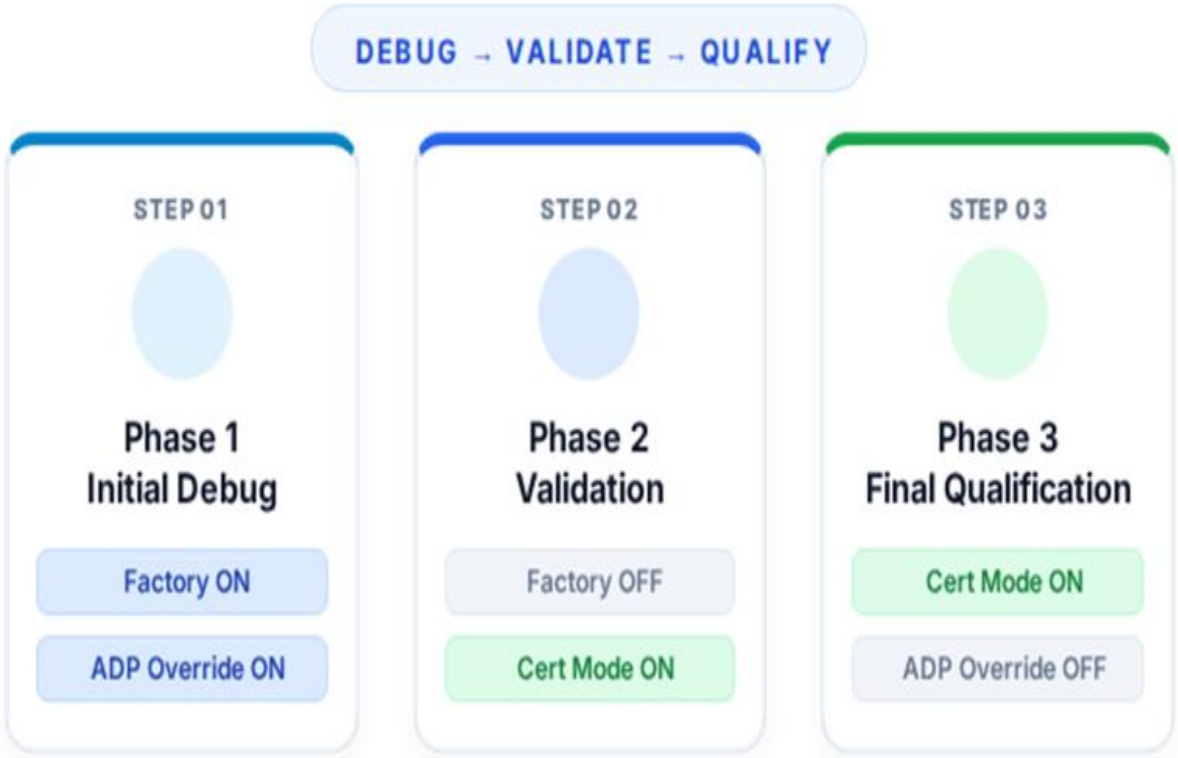
ADP Optional File Management

- Certification Mode
  - Turns off certain miscellaneous Test Plan requests
  - NS must fill the drive; Firmware Rev requirements
- Factory Mode
  - Runs as “Certification Mode” but turns off all preconditioning
- ADP Override Mode
  - Bare Minimum to function; Test will move forward w/o checking parameter information that is supposed to be entered per every drive serial number and firmware version



ADP Optional File Management

- Phase 1 - Initial Debug
  - Focus on major issues
  - Factory Mode Enabled w/ADP Override Enabled
- Phase 2- Initial Debug Validation
  - Disable Factory Mode
  - Run with Certification Mode Enabled
- Phase 3- Final Qualification
  - Test Run with Certification Mod Enabled
  - ADP Override disabled



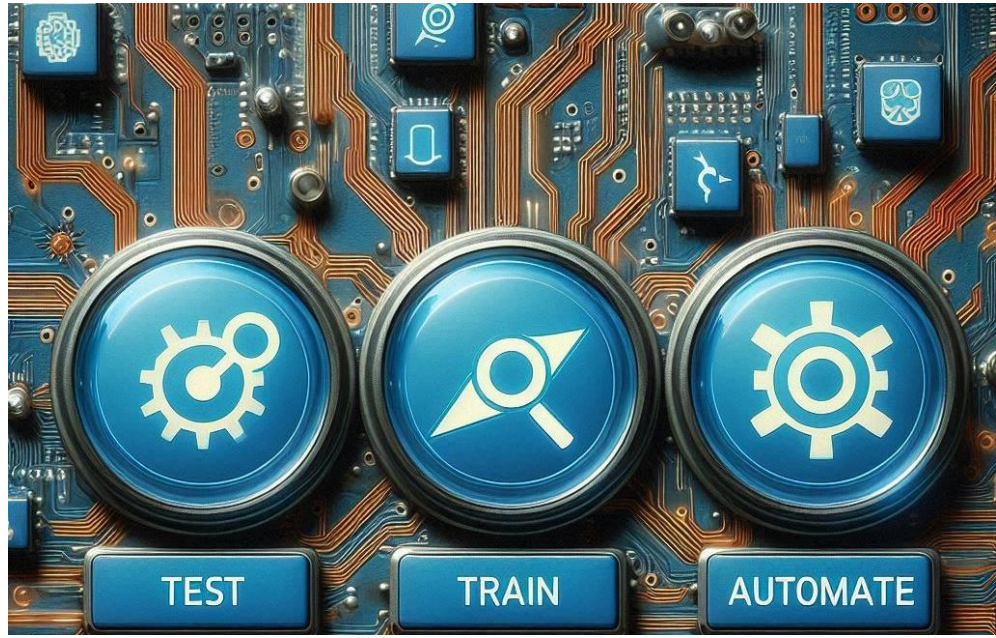
SANBlaze Call to Action

- Drive SSD Standard Benchmark
  - Let's Create a formalized Certification Program?
- PCI-SIG/NVMe Conformance
- Drive Industry Support
- Tighten Standards and Consistency of deliverables
- Where to find additional information (URL links)
  - https://www.sanblaze .com; http://unh.iol.edu
- OCP Market Place OCP.org
- UNH/IOL Test services-OCP testing on SANBlaze
  - Lab environment test/OCP Experience Lab

TELEDYNE LECROY
Everywhereyoulook™

Austin Labs
TESTING TRAINING

- Third Party Testing to provide OCP Compliance as a Service
- In-depth Protocol training for NVMe and PCIe specifications
- OCP Compliance test SW for NVMe SSD's



Broad array of test solutions for serial data standards and next-gen technologies

Austin Labs OCP Experience Center

OCP Experience Centers seed emerging markets by allowing potential adopters to see and understand OCP equipment, channel and system integrators to conduct presales and engineers to develop new pre-product concepts.

With state-of-the-art test facilities located around the world and industry expertise Austin Labs takes advantage of the wide array of Teledyne LeCroy tools for validation testing of products from server/storage to client systems.

Come test with Austin Labs and improve your products!



OCP Experience Center

Austin, TX, USA

OCP Inspired Solutions

OCP Inspired Test Tools

OakGate SSD Appliances are designed to support OCP NVMe SSD qualification across varying form factors plus allows for complete control of performance, power management and LED specific tests.



OCP Datacenter NVMe SSD Specification Compliance Suite

Built to quickly help validate NVMe SSD drives against the latest specifications.



OakGate SSD Appliances

OakGate SSD Appliances are designed to support OCP NVMe SSD qualification across varying form factors plus allows for complete control of performance, power management and LED specific tests.



OCP PCIe 6.0 OCP 3.0 NIC Interposer

The PCIe 6.0 OCP NIC 3.0 Interposer allows users to connect a Teledyne LeCroy PCIe protocol analyzer between an OCP NIC 3.0 device and an OCP Server System to monitor, capture, record and analyze protocol traffic.

OCP Testing Platform

- Desktop appliance or rackmount solution
- Capacities up to 48 SSDs
- PCIe 4.0, 5.0, and 6.0
- **Complete PCIe 6.0 solution Platform and Modules**
- U.2, M.2, U.3, CEM, EDSFF E1.S or EDSFF E3.S/L SSDs
- Traffic Generation and Error Injection
- **Full I3C Support**
- **Power Measurements and Cycling**



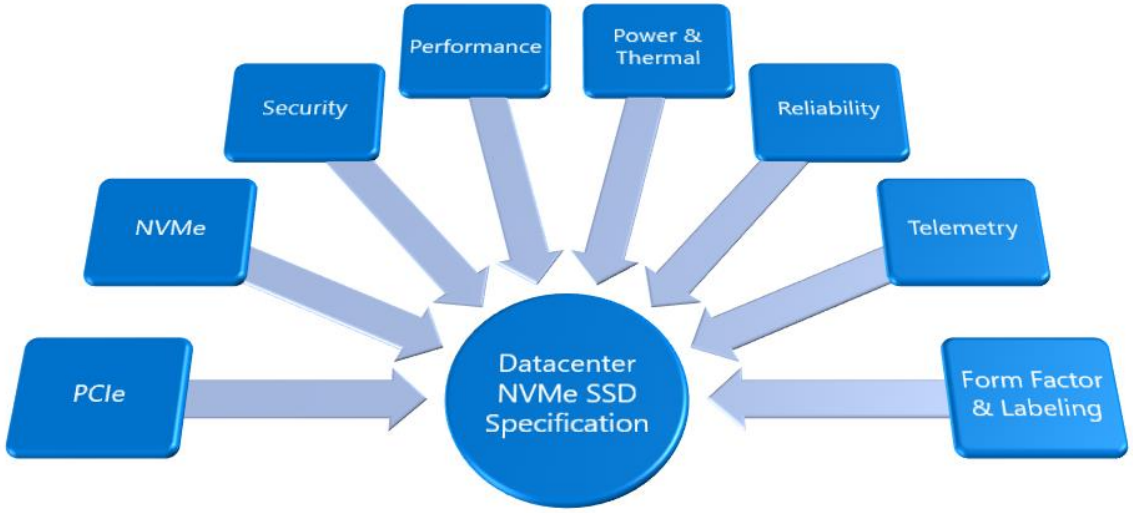
OCP Compliance Test Tool Suite

- Simple Easy to use GUI
- Provides Pass/Fail Results
- Power Control
- Full Logging and Debug
- Validates qualification to spec
- Tested and approved by Microsoft for CDL (Certified Device List)
- Ability to add customized tests to support other OCP gap requirements
- Available: OCP Marketplace



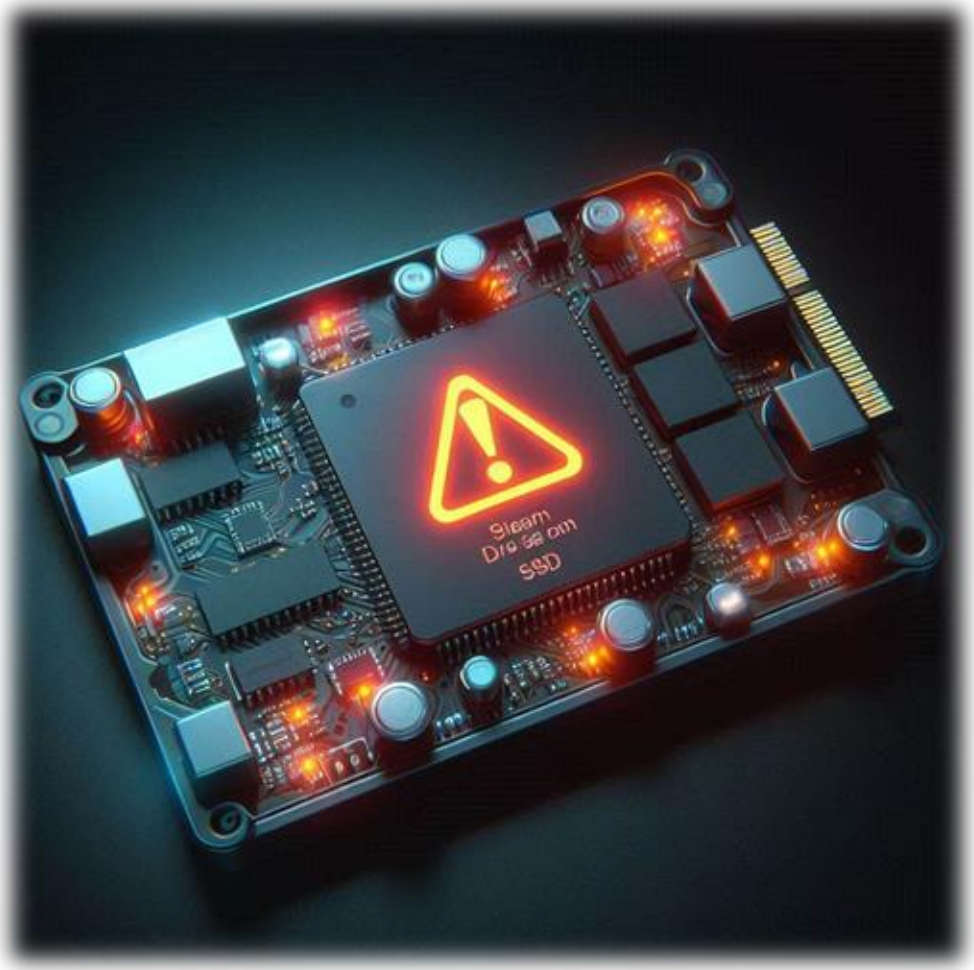
What is the OCP Compliance Test Suite

- Collaborative Specification
- Driven by Microsoft, Meta, Dell, HPE, Google
- Complimentary to NVMe Specifications
- Aligns SSD Requirements between OEMs and SSD companies
- Drives test development and automation
- Enables focus on “customer” requirements
- More efficient SSD Development – saves money
- Enables 3rd party test development
- Aligns drive vendors on specific cloud-based SKU’s
- More Features, Better Quality, and Faster Delivery



What is new for OCP

- Security
- TCG
- Sanitize
- Debuggability
- Smart Logs
- Health Information
- Latency Monitor
- Telemetry Profile
- Asynchronous Event Requests



Where can I find more information?

- Teledyne LeCroy - All OCP Solutions

https://www.teledynelecroy.com/protocolanalyzer/ocp.aspx

- OCP Marketplace

https://www.opencompute.org/products?cloud_products%5Bquery%5D=lecroy

- Austin Labs

https://www.teledynelecroy.com/services/austinlabs-testing.aspx

- Oakgate Technologies

https://www.teledynelecroy.com/ssdtesting/

- OCP Experience Center

https://www.opencompute.org/facilities/56/ocp-experience-center-austin-tx-hosted-by-teledyne-lecroy





Call to Action - Collaboration

- OCP Specification is continuing to grow
  - Connect
    - It takes all of us from the OEM/Hyperscale companies to the SSD suppliers to the tool manufacturers for success
  - Collaborate
    - I am inviting you to be a part of the test requirements
      - We improve our products through you!
  - Accelerate
    - We are working together to solve the challenges for tomorrow

OPEN Compute Project®

Connect

Collaborate

Accelerate

Thank You!



Open Discussion

